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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

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1. Team & Project Overview

Team Name	Haze&Dust
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8, SDG 9, SDG 12
Project / Solution Name	Vyte
Name (Leader)	Yau Zen Hao
Email & Phone (Leader)	Email : zenhao5514@gmail.com Phone : 010 863 3226
University Affiliation(s)	Tunku Abdul Rahman University of Management and Technology (TARUMT)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysia has 1,069,831 MSME establishments (DOSM, Economic Census 2023). Many are small retailers, F&B outlets, pharmacies and B2B distributors that still track stock, supplier quotes, orders, invoices and renewals with paper, WhatsApp messages and spreadsheets. We are tackling three connected problems they face every day.

First, inventory waste and supplier defect losses. Without expiry tracking or supplier defect memory, perishable stock expires on the shelf, defective supplier batches slip through and slow-moving items tie up cash. Worldwide, about 13% of food is lost between harvest and retail and a further 19% is wasted at retail, food-service and household level (UNEP Food Waste Index Report 2024).

Second, manual admin and cash-flow blind spots. Retyping multilingual WhatsApp orders and supplier invoices causes errors, overselling and late payments, and owners cannot see stock holding costs or the minimum safe cash buffer needed to survive their sell-off cycle.

Third, missed compliance. Malaysian RMCD SST tax coding (0%, 6%, 8%, 10%) and renewals for premise licences, shop tenancy, commercial fleet road tax/PUSPAKOM, OEM warranties and software subscriptions are tracked from memory, leading to penalties and lapsed contracts.

These problems affect SME owners, their staff, suppliers and customers. Economically, waste and errors squeeze thin margins and working capital. Environmentally, expired stock ends up as waste. Socially, struggling small businesses put local jobs and livelihoods at risk. Full ERP systems can solve this but are too costly and complex for most SMEs, so they need a simple, affordable tool.

(246 words)

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Vyte is an AI-powered operational hub that converts raw order messages, receipt photos, and legacy files into structured inventory tracking, automated restock insights, expiry & quality alerts, and cash-flow runway calculations. It applies AI across six core modules and system-wide governance tools:

1. Auto-Restock And AI Supplier Bidding:
    - Tracks F&B, Retail, and B2B stock against minimum thresholds, scores supplier RFQ bids using price, historical defect memory, and staff sample tests, and preserves historical financials for decommissioned SKUs.

2. Multilingual NLP Order Intake:
    - Parses WhatsApp/email orders (English, Manglish, Bahasa Melayu, Mandarin), photos, and CSVs against catalog SKUs, drafts AI WhatsApp replies, locks orders when stock is low, and transfers shortage quantities to Restock.

3. Sales Analytics And Cash-Flow Runway:
    - Computes SKU depletion velocity, time-to-sell-off, holding costs (1–50%), 21-day turnover restock recommendations, and working-capital runway.

4. Batch Quality And Expiry Control:
    - Scans perishables (<5 days left) to recommend combo promos or quarantine, assigns tasks to Orbyte HR warehouse staff with SLA due dates, and auto-escalates overdue tasks.

5. Smart Invoices And RMCD SST Engine:
    - Extracts receipts via AI OCR/CSV, classifies expenses, computes Malaysian SST (0%/6%/8%/10%) with SST-02 CSV export, models 30-day minimum safe cash (+20% buffer), and sends EN/BM WhatsApp payment reminders.

6. 5-Pillar Compliance Vault And Governance:
    - Monitors Tenancy, Fleet (JPJ/PUSPAKOM), OEM Warranties, Software SaaS, and SSM/PBT licences with a Document Vault, Concierge/RPA filing, Immutable Audit Logs, an AI Universal System Migrator (Shadow/Copilot/Autopilot), 4-currency live FX (MYR/SGD/USD/RMB), 24/7 Support, and RBAC.

Vyte operates as a connected intelligence plug-in for Orbyte's ecosystem which emits standardized operational payloads into Orbyte's workforce webhooks without replicating HR functions and leverages our SDC Phase 1 workshop skills in GenAI/NLP APIs, RESTful routes, SQLite, and RunPod GPU inference.

(292 words)

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend:
- TypeScript, React 19 and Vite: frontend components, UI and build tooling
- Tailwind CSS and Lucide React: web styling, responsive layout and page structure

Backend & Design:
- Kotlin (JVM 21) REST API built with Gradle: inventory, invoice, quality-control and audit-log endpoints
- Figma: UI/UX wireframing and prototype workflows

AI & External APIs:
- Google AI Studio Gemini API (own account): multilingual (EN/Manglish/BM/Mandarin) order parsing, invoice field extraction and structured JSON output.
- RunPod Serverless GPU endpoint (sponsor credits): GPU inference for supplier defect-risk scoring, holding-cost forecasting, and expiry mitigation analysis
- Public exchange-rate APIs (ExchangeRate-API, Frankfurter, CDN Currency API): live MYR, SGD, USD and RMB conversion
-WhatsApp Click-to-Chat API: 1-click dispatch for supplier POs, multilingual AI customer replies, overdue AR reminders, and 24/7 engineer support.

Database & Hosting:
- SQLite (vyte.db): products, suppliers, approval tasks, invoices and audit logs
- Replit (sponsor-provided): development environment and live prototype hosting; GitHub for source control

React UI → Kotlin REST API ↔ Gemini API / RunPod GPU → Owner/Manager approval → SQLite database → results and audit log shown in the UI

Yes. We used two SDC sponsor resources. RunPod GPU credits (RunPod Serverless) run AI model inference for Vyte's analysis features; the Kotlin backend calls the RunPod endpoint and returns structured JSON to the application. Replit serves as our development environment and hosts the live prototype. We also use the Google AI Studio Gemini API through our own account (not a sponsor resource) for order parsing and invoice extraction.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied the five-stage Design Thinking method to make sure Vyte solves real user pain points and is ready for real deployment by Phase 2:

- Empathise: identified SME pain points — manual order entry, expired and dead stock, tedious invoice sorting and missed renewal deadlines
- Define: focused on high-impact features, such as restock alerts, holding-cost analysis, expiry tracking, invoice extraction and renewal tracking
- Ideate: for user-friendliness, designed a simple card-based interface, an Interactive Spotlight Tutorial on first login, Universal System Migrator modes (Shadow/Copilot/Autopilot), and a Human-in-the-Loop RBAC checkpoint so an Owner or Manager verifies AI results before the database changes
- Prototype: for technical viability, built a React frontend, a Kotlin REST API, Gemini and RunPod AI integration and a SQLite database
- Test and Deploy: tested end-to-end workflows (stockout locks, MOQ adherence, decommissioned SKU history, and overdue QC escalation payloads), added a mock mode for offline demos plus a 24/7 After-Sales Support Portal, and deployed the prototype on Replit, ready for Phase 2

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 12.3: By 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains. Target 12.5: By 2030, substantially reduce waste generation through prevention, reduction, recycling, and reuse. Target 8.2 & 8.3: Achieve higher levels of economic productivity through technological upgrading and support the formalization and growth of micro-, small- and medium-sized enterprises (MSMEs). Target 9.3: Increase the access of small-scale industrial and other enterprises to financial services, affordable credit, and value-chain integration.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Malaysia has 1,069,831 MSME establishments (DOSM, Economic Census 2023), most of them small retailers, F&B outlets, grocers, pharmacies and distributors. Many still run their daily operations on paper records, WhatsApp messages and spreadsheets. This creates the following problem: 1. Perishable & Inventory Waste Baselines: Globally, 13% of food is lost in supply chains and 19% is wasted at retail and food-service levels (UNEP Food Waste Index Report 2024). In Malaysia, unoptimized inventory management causes retail and F&B SMEs to lose an estimated 35 to 45 kg of perishable stock

Metric Category	Your Response
	(equivalent to RM 1,200 – RM 1,800 in inventory value) per outlet each month due to untracked shelf-life expiration, unvetted supplier batch defect rates (8%–11%), and over-purchasing (SWCorp Malaysia & DOSM MSME Waste Benchmarks) 2. Manual Operational Hours Baseline: Malaysian MSME operators spend an average of 14 hours per week manually retyping multilingual WhatsApp orders, cross-checking physical stock, sorting receipts for SST filing, and tracking license renewals across disconnected spreadsheets (DOSM Economic Census 2023 / SME Corp Digitalization Survey) 3. Statutory & Contract Compliance Baseline: An estimated 30%–35% of MSMEs miss critical annual deadlines for local council premise licences, commercial vehicle road tax/PUSPAKOM, or tenancy contract options, incurring penalty surcharges and forfeited multi-year discounts (SME Corp Digitalization Survey / PBT Municipal Records).
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1. Kilograms (kg) and Ringgit (RM) value of expired perishable inventory and defective supplier stock wasted per month 2. Hours spent per week on manual order processing, SST invoice categorization, and compliance renewal preparation 3. Percentage (%) of licence and compliance renewals (premise licences, tenancy agreements, road tax and PUSPAKOM, warranties and software subscriptions) completed before their expiry date per quarter

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation e.g., "20% improvement in computational resource use."*	1. 40% reduction in monthly perishable food and defective inventory waste (salvaging approximately 15–18 kg or RM 500–RM 720 of stock per outlet monthly) achieved through Vyte's automated <5-day shelf-life discount bundle triggers and defect-scored supplier RFQ bidding. 2. 60% reduction in weekly manual administrative hours (saving ~8.5 hours per week per business) through multilingual NLP order intake, automated RMCD SST-02 expense classification, and centralized 5-pillar renewal checklists. 3. 95% on-time renewal rate across premise licences, tenancy agreements, vehicle fleet road taxes, and equipment warranties, eliminating statutory late penalties and securing up to 22% multi-year bundle savings.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React	Code (Framework and Build Tool)	MIT License
TypeScript	Code (Language)	Apache 2.0
Kotlin (JVM 21) & Gradle	Code (Backend Language & Build Tool)	Apache 2.0
Tailwind CSS & Lucide React	Code (UI Styling & Icons)	MIT / ISC License
SQLite	Code (Database)	Open Source
Gemini API	Pre-trained AI Model	Google Terms of Service
Claude Sonnet	Pre-trained AI Model	Anthropic Terms
RunPod Inference	Pre-trained AI Model	RunPod Terms

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



24-10-2026